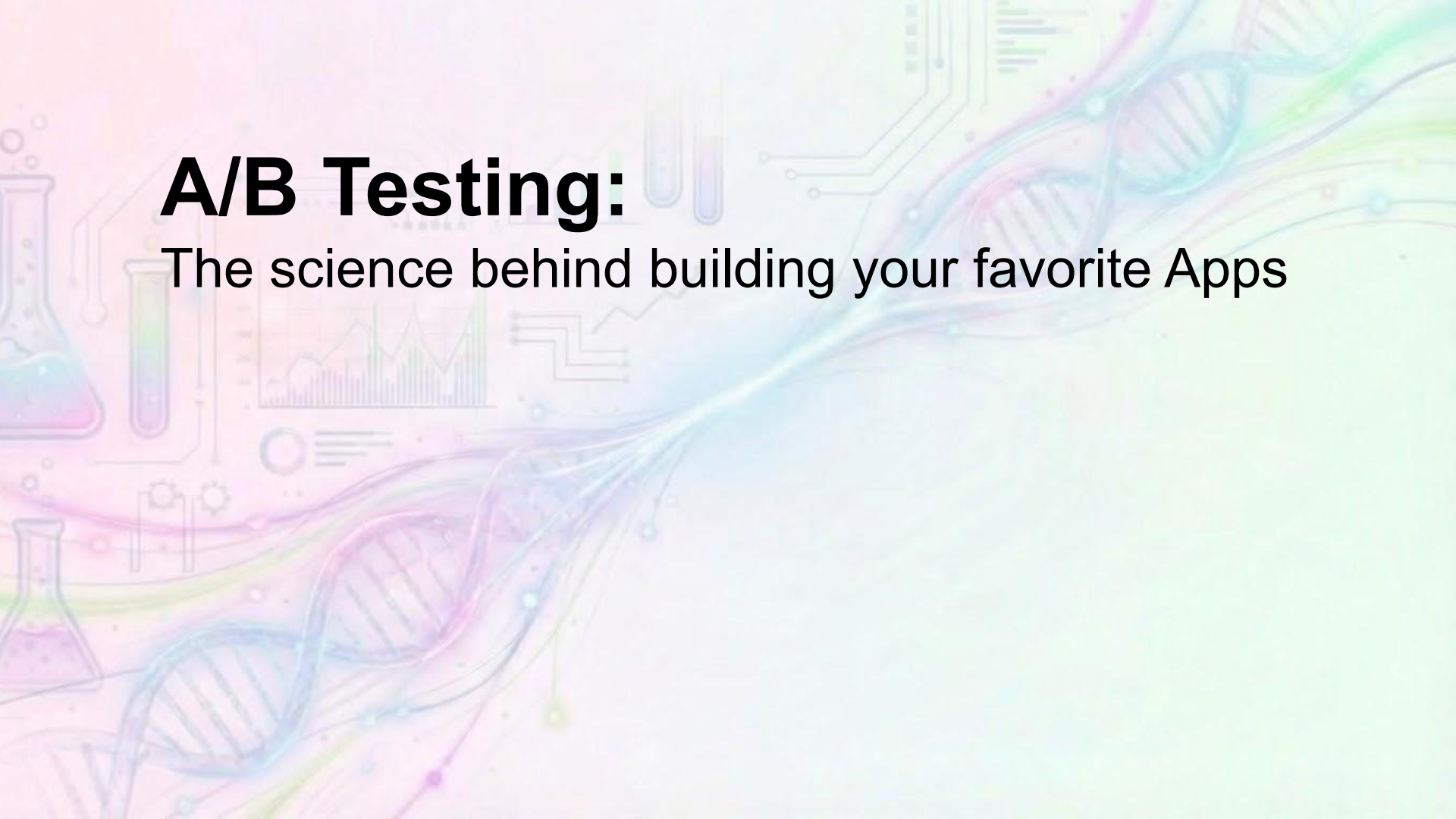




A/B Testing:

The science behind building your favorite Apps

- 
- Why does your Instagram feed look different from your friend's?
 - Why is the "Sign Up" button blue for you but green for them?
 - Why do you have some features but not your friends?

Answer: A/B testing

- Tech companies are running thousands of A/B tests to understand scientifically what works
- People are randomly assigned to test and control groups to understand which group performs better

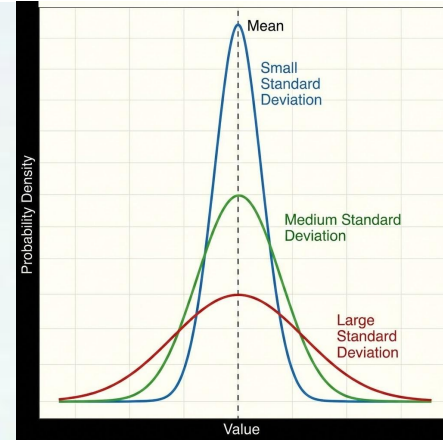
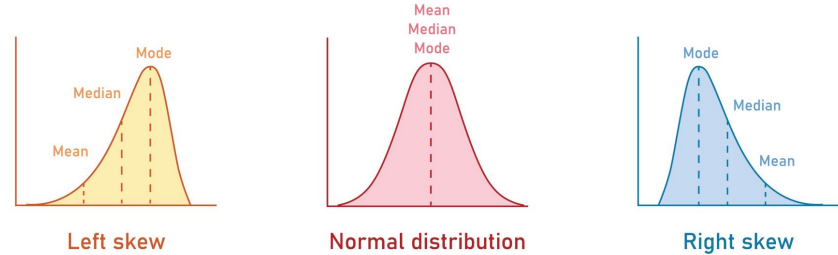
What are A/B Tests?

Let's revise some basics of statistics

Measures of Central Tendency

- Mean: Average
- Median: Middle point
- Mode: Most Frequent
- Standard Deviation: How spread out is the distribution

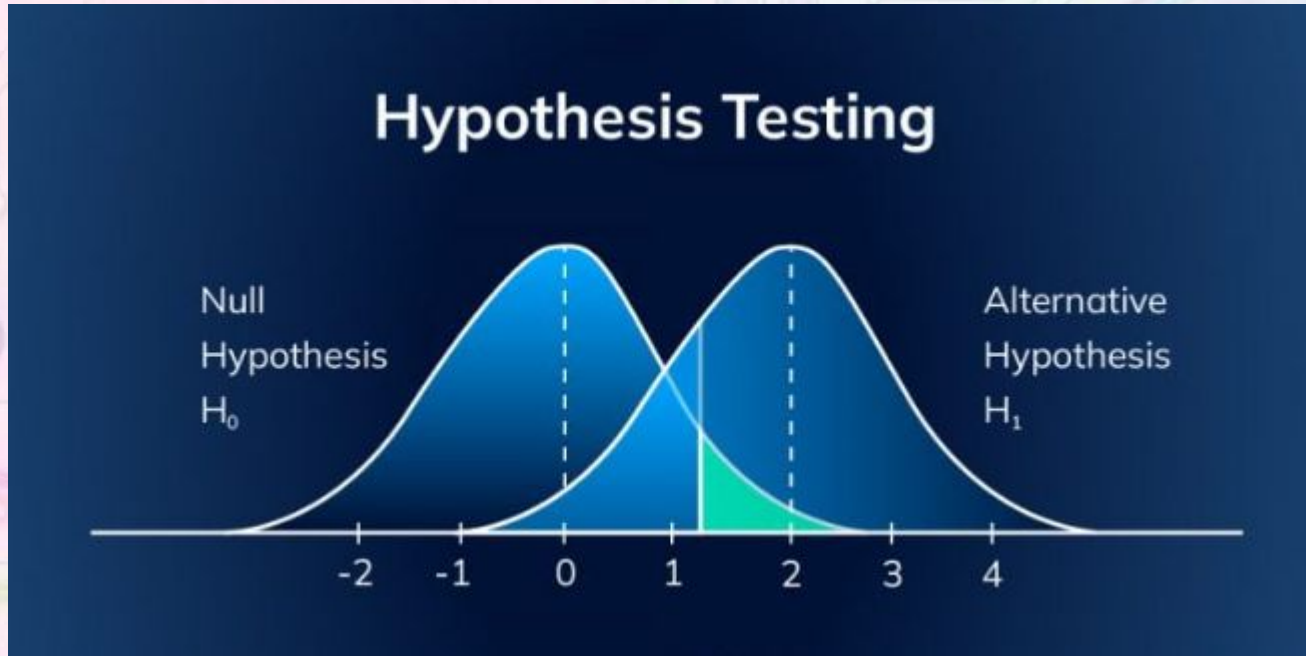
Mean, Median and Mode



Hypothesis Testing

- **The Null Hypothesis (H0):** "The change we made did absolutely nothing."
- **The Alternative Hypothesis (H1):** "The new feature increased user engagement."
- **P-Value (The "Luck" Factor):** If $p < 0.05$, there is less than a 5% chance the result was just a fluke. In tech, we want to be sure it wasn't just luck

Hypothesis Testing



A/B Testing – The Gold Standard

- **What is it?** A randomized controlled trial for software.
- **The Split:** 50% see the old (Control), 50% see the new (Treatment).
- **Why Randomize?** It ensures both groups have the same mix of ages, locations, and interests.
- **The Outcome:** Any difference in behavior **must** be caused by the change.

A/B Testing in Big Tech (Deep Dive)

- **Scale:** Google runs roughly **25,000+ experiments** per year. You are likely in 50+ experiments right now.
- **Bucketing:** How do we keep users in the same group? (Using "User IDs" or "Cookies").
- **Network Effects:** If you test a "Like" button on Instagram, it affects both the liker and the poster. Tech firms use **Graph Cluster Randomization** to handle this.
- **Success Metrics:** Metrics to improve
- **Guardrail Metrics:** Metrics that should not deteriorate

Holdouts – The "Long Game"

- **The Problem:** Small wins can lead to a big mess. If every team adds one "small" pop-up, the app becomes unusable.
- **The Solution:** Keep 1% of users on the "Version from 6 months ago."
- **The Insight:** This allows us to see the **cumulative effect** of 100 different A/B tests.
- **Speaker Note:** Think of it like a "Time Capsule" group.

Causal Inference: Correlation $\neq$ Causation

- **The Trap:** People who use "Premium" features might stay on the app longer. Is it because the features are good (Causation), or because they were already "Super Users" to begin with (Correlation)?
- **The Science:** Using math to simulate an experiment when we can't run one.
- **The "Counterfactual":** Asking "What would have happened to this user if they *hadn't* seen the ad?"

Guardrails & Ethics

- **Guardrail Metrics:** We want to increase "Time Spent," but not at the cost of "App Crashes" or "Uninstalls."
- **Ethics:** Is it okay to test things that make an app more "addictive"?
- **Responsibility:** Data Scientists are the "Ethical Gatekeepers" of the product.

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Conclusion

- **Summary:** Data Science is a combination of Statistics and Machine Learning.

Statistics (A/B testings) is an important part of Data Science

Next time an app updates, ask yourself: *"Am I the Treatment or the Control?"* :)



Questions?